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ORIGINAL RESEARCH

Evening Alpha-Lactalbumin Supplementation Alters Sleep Architecture and Reduces Morning Reaction Time in an Athletically Trained Population With Sleep Difficulties

Jackson Barnard,[1] Spencer Roberts,[1] Michele Lastella,[2,3] Damien L. Callahan,[4] Brad Aisbett,[5] and Dominique Condo[1]

[1]Centre for Sport Research, Institute of Physical Activity and Nutrition, Deakin University, Geelong, Australia; [2]Appleton Institute for Behavioural Science, Central Queensland University, Wayville, Australia; [3]School of Health, Medical and Applied Sciences, Central Queensland University, Rockhampton, Australia; [4]School of Environmental and Life Sciences, Burwood Campus, Deakin University, Geelong, Australia; [5]Institute for Physical Activity and Nutrition, Deakin University, Geelong, Australia

Evening consumption of a whey protein rich in the amino acid tryptophan, alpha-lactalbumin (ALAC), has previously shown to benefit sleep—particularly among poor sleepers. Given trained populations often experience sleep difficulty, this study investigated whether evening supplementation of ALAC would influence sleep outcomes, mood, and next-day cognitive performance within a trained population with sleep difficulties. Nineteen trained participants (females, $n = 11$) with sleep difficulties (Athlete Sleep Screening Questionnaire: 8.1 ± 3.1; Pittsburgh Sleep Quality Index: 10.5 ± 4.1) completed this double-blinded, counterbalanced, randomized, crossover trial. Forty grams of ALAC or control were supplemented 2 hr presleep for three consecutive nights in a controlled environment, with sleep measured using dry electroencephalography. Blood samples were taken on the first evening of each experimental trial, with mood, sleepiness, and recovery assessed across the evening and morning. A cognitive testing battery was also completed each morning. During the ALAC condition, the primary findings were that participants had raised plasma tryptophan levels ($p < .01$), increased nonrapid eye movement Stage 2 sleep duration (CON: 205.9 ± 33.3; ALAC: 216.5 ± 33.1 min), reduced rapid eye movement duration (CON: 110.8 ± 27.9; ALAC: 99.7 ± 23.1 min), and improved reaction time in cognitive tests involving sensory motor speed, spatial orientation, and vigilant attention ($p < .05$). Data suggest evening supplementation of 40 g ALAC alters sleep architecture and improves next-morning reaction time in trained populations with sleep difficulties. Therefore, trained individuals experiencing sleep difficulty may benefit from acute ALAC supplementation to assist next-day performance. Future research should investigate this effect within habitual environments, outside of a tightly controlled setting.

Keywords: diet, protein, sleep, whey

Athletes consider sleep an important behavior, with inadequate sleep influencing recovery, performance, and overall wellness of an athlete (Walsh et al., 2021). While some studies suggest athletes obtain healthy sleep (Litwic-Kaminska & Kotysko, 2020), a recent expert consensus stated that many do not, with up to 78% of athletes experiencing sleep difficulties (Walsh et al., 2021). Factors such as precompetition anxiety, competition scheduling, and high training loads can disturb athletes' sleep (Roberts et al., 2019), potentially leading to worse sleep than their nonathlete counterparts (Demirel, 2016; Leeder et al., 2012). With cumulative sleep debt impairing many outcomes including cognition, mood, recovery, and sports performance (Knowles et al., 2022; Walker, 2009; Walsh et al., 2021), effective practical strategies to alleviate sleep difficulties within athletes remain warranted.

The influence of tryptophan (TRP) rich foods on sleep is of growing interest (Barnard et al., 2024). TRP is an aromatic amino acid that is a dietary precursor to serotonin and melatonin, both of which promote sleep through sleep initiation and maintenance (Richard et al., 2009). As TRP competes against other large neutral amino acids (LNAA) for transport across the blood–brain barrier; the ratio of TRP:LNAA within the blood affects TRP availability to the brain and, thus, melatonin production via the serotonin pathway (Richard et al., 2009). The richest source of TRP found within the diet, α-lactalbumin (48 mg TRP per 1 g protein), raises the TRP:LNAA ratio by ~130% when supplemented at 40 g, which is significantly greater than the 40%–50% increase believed necessary to augment brain TRP availability and serotonin synthesis (Markus et al., 2005).

Supplementation of 20–40 g α-lactalbumin (≤3.5 hr prior to sleep) has been shown to reduce sleep latency and promote sleep quality, particularly within populations experiencing sleep difficulties (Barnard et al., 2024). Unlike other sleep hypnotics, α-lactalbumin appears to improve morning alertness and sleepiness (Markus et al., 2005) and may promote next-day sporting performance (Miles et al., 2021). With sleep-promoting strategies such as maintaining usual sleep/wake times and trialing exogenous melatonin not always feasible to athletes, α-lactalbumin supplementation could be a practical strategy to mitigate sleep difficulty within athletes.

In athletic populations, the impact of evening α-lactalbumin supplementation has been investigated in four studies (Gratwicke et al., 2021; MacInnis et al., 2020; Miles et al., 2021; Oikawa et al., 2019). Two studies in trained females observed sleep changes following supplementation, including reduced sleep latency via

Roberts https://orcid.org/0000-0002-9095-7070
Lastella https://orcid.org/0000-0003-1793-3811
Callahan https://orcid.org/0000-0002-6384-8717
Aisbett https://orcid.org/0000-0001-8077-0272
Condo https://orcid.org/0000-0002-8348-7488
Barnard (jgbarnard@deakin.edu.au) is corresponding author, https://orcid.org/0000-0001-5160-1728

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actigraphy (Gratwicke et al., 2021) and an increased polysomnography-derived nonrapid eye movement (NREM) Stage 2 proportion (Miles et al., 2021), whereas two studies in endurance-trained males and females had no significant sleep findings (MacInnis et al., 2020; Oikawa et al., 2019). Limitations of these studies necessitates further research that controls for diet, recruits poor sleeping populations, and uses comprehensive sleep analysis tools (i.e., electroencephalography [EEG]). For instance, most dietary protein sources contain high amounts of LNAA and low TRP, meaning consumption of mixed-meals close to α-lactalbumin supplementation can blunt the TRP:LNAA response—with three studies not reporting evening meal timings (Gratwicke et al., 2021; MacInnis et al., 2020; Oikawa et al., 2019). Further, null findings were observed in two actigraphy-based studies where participants averaged short sleep latencies (<3 min) and high sleep efficiencies (>87%) throughout the trial (MacInnis et al., 2020; Oikawa et al., 2019), limiting the scope for sleep improvements. With only one athlete study having measured the influence of α-lactalbumin on sleep staging, more extensive research is required. Therefore, the purpose of this study was to determine the efficacy of 40 g α-lactalbumin supplementation across 3 days on sleep architecture, mood, and cognition in a trained population with sleep difficulties.

Methods

Participants

Nineteen trained participants (females, $n = 11$) with sleep difficulties (Athlete Sleep Screening Questionnaire: 8.1 ± 3.1; Pittsburgh Sleep Quality Index: 10.5 ± 4.1) participated in the study (Table 1). Sample size was calculated using PASS software (NCSS) with sleep onset latency as the primary outcome. A two-sided *t* test achieved 90% power when the total sample size is 18, the square root of the within mean square error is 0.5, and the actual mean difference is −0.6 (Halson et al., 2020; Miles et al., 2021). Participants were required to be exercising ≥3 days per week for a minimum total of ≥5 hr per week to be identified as *trained* (De Pauw et al., 2013), with most participants completing *resistance exercise* and/or *running*. Inclusion criteria required participants to have mild or worse sleep difficulty (score ≥ 5) on the Athlete Sleep Screening Questionnaire, poor sleep quality (global score > 5) on the Pittsburgh Sleep Quality Index, and self-reported habitual sleep latency as ≥15 min on the Pittsburgh Sleep Quality Index. Exclusion criteria included the following: sleep medication use, smoking, excessive alcohol consumption (>17 standard drinks per week), dairy allergy, high caffeine use (e.g., >5 $mg \cdot kg^{-1} \cdot d^{-1}$), antidepressant medication, diagnosed sleep disorder (e.g., sleep apnea), current or recently finished night shift work, and transmeridian travel within the past month. This study was approved by the Deakin University Human Research Ethics Committee (2021-372) and conducted in accordance with the Declaration of Helsinki.

Table 1 Participant Demographic and Characteristic Details

Variable	Mean ± *SD*
Age (years)	25.5 ± 5.6
Body mass (kg)	75.7 ± 13.0
Height (m)	1.7 ± 0.1
ASSQ (SDS)	8.1 ± 3.1
PSQI (global score)	10.5 ± 4.1
Subjective sleep latency (min)	30.2 ± 15.7

Note. ASSQ = Athlete Sleep Screening Questionnaire; PSQI = Pittsburgh Sleep Quality Index; SDS = sleep difficulty score.

Study Design

A double-blinded, randomized control, counterbalanced, crossover experiment required participants to attend testing facilities on seven occasions (Figure 1). Prior to commencing the experimental trials, an overnight familiarization was completed, where participants performed cognitive testing and slept within the testing environment. Postfamiliarization, participants completed five consecutive days of electronic-based food and sleep diaries, allowing researchers to individually standardize diet and sleep/wake times based on individual habitual averages. This ensured that a participant's habitual dietary and sleep patterns were replicated, despite being in a laboratory-based environment.

Experimental Trials

For three consecutive nights, participants resided within research accommodation at Deakin University, with habitual activities continued throughout the daytime. On the first night, participants arrived at least 4 hr prior to sleep with a standardized low-protein dinner (0.2 g/kg bodyweight) provided at this time. An indwelling cannula was inserted into a forearm vein 2.5 hr before individually prescribed bedtimes. Two hours prior to sleep, participants consumed a premixed protein drink within 10 min, containing either 40 g α-lactalbumin or 40 g collagen protein (control). Blood samples were drawn immediately prior to consumption then again at 30-, 60-, 90-, and 120-min postconsumption. Subjective sleepiness was self-rated using the 9-point Karolinska Sleepiness Scale (KSS) and mood via the Brunel Mood Scale (BRUMS) by participants every half an hour from supplement ingestion until sleep. Additionally, the Short Recovery Stress Scale in Sport (SRSS) was completed at 90 min postsupplement ingestion. The cannula was removed following the final blood sample at the prescribed bedtime, with participants then getting ready for sleep. Once ready for sleep, a dry-EEG headband was fitted. The sleep recording started when participants were positioned in bed ready to initiate sleep.

At the individually prescribed wake time, participants were awoken. Thirty minutes after waking, participants completed the KSS, BRUMS, SRSS, and sleep diary. Following completion of these surveys, participants completed a cognitive testing battery via a tablet (Apple iPad; Apple) within a silent room and were then provided with standardized meals for the day, with instruction to return to the facility at least 3 hr prior to bedtime. Supplementation and survey timelines were replicated on Nights 2 and 3; however, no cannulation or blood sampling was required.

After completion of cognitive testing on the morning of Day 4, participants returned to their usual residence for a minimum 5-day washout period where no testing or supplementation occurred. Following the washout period, participants arrived back to Deakin University at least 4 hr prior to sleep, where methods were replicated for the next three nights, receiving the opposing supplement for the second trial.

Living Conditions

Participants were housed within the Deakin University Burwood sleep laboratory (Knowles et al., 2022). A maximum of three participants were accommodated at any one time, with each

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participant having their own room fitted with a single adjustable bed. In the evening, participants were to turn off blue-light emissions on electronic devices at least 2 hr before sleep and cease electronic device use 1 hr prior to sleep. Temperature was controlled between 18 and 20 °C within each room, with access to bathroom and kitchen facilities external to the rooms. Participants were not permitted to sleep outside of prescribed hours (i.e., napping), which was verbally confirmed each evening.

Dietary Supplement

Sequence of supplementation for the crossover was allocated via simple randomization in a counterbalanced manner by an external researcher. The experimental supplement consisted of 40 g α-lactalbumin (BiPRO Alpha 9000; Agropur Inc.), and the control contained 40 g collagen (Collagen Hydrolyzed Protein; True Protein), with the amino acid content of each supplement provided in Table 2. Four hundred milliliters of water was added to the supplement, with taste and texture matched using 15 g sugar-free chocolate powder (Avalanche) and 4 g Stevia (New Zealand Sugar Company) (Miles et al., 2021). Collagen was selected as the control supplement as it satisfied four primary criteria: minimal impact on the primary outcome of objective sleep latency (Thomas et al., 2024), isocaloric, protein-based, and low in TRP and other LNAAs (Table 2). By using a protein-based control, we recognize that collagen may not be a completely inert substance, whereby improvements in next-morning cognition and awakenings have been observed previously within populations with sleep complaints

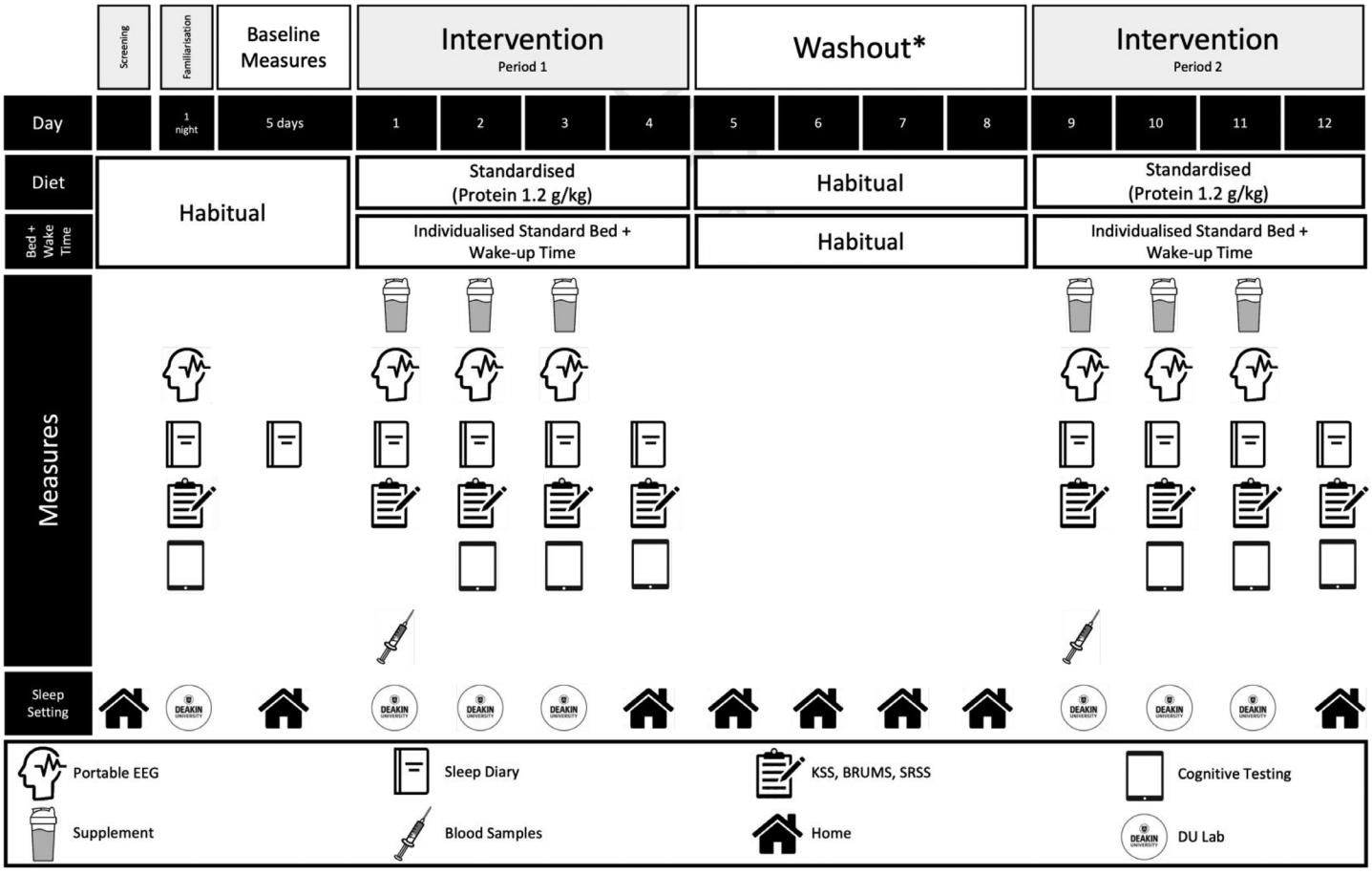

Figure 1 — Graphic overview and timeline of the study. DU Lab = Deakin University laboratory; KSS =Karolinska Sleepiness Scale; SRSS = Short Recovery Stress Scale in Sport; EEG = electroencephalography; BRUMS = Brunel Mood Scale. *Minimum washout, with naturally menstruating females requiring an extended washout period.

Table 2 Large Neutral Amino Acid Composition per 40 g of Protein Supplements

Large neutral amino acids	α-lactalbumin (40 g)	Control—collagen (40 g)
Isoleucine (g)	2.4	0.5
Leucine (g)	4.3	1.0
Phenylalanine (g)	1.6	0.6
Tyrosine (g)	1.8	0.2
Valine (g)	1.7	0.9
Tryptophan (g)	1.9	0.0

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(Thomas et al., 2024). We do, however, note that the magnitude of difference with potential cognitive improvements may be greater if comparing to an inert substance.

Dietary Standardization

Participants were provided all meals (i.e., two breakfasts, two lunches, and three dinners) and snacks throughout the trial using an external macronutrient-controlled meal provider (eXplosive Custom Macros). All meals were individually matched for calories in line with the 5-day food diaries provided postfamiliarization via the smartphone application Easy Diet Diary (Xyris Software). To limit the potential confounding effect of dietary protein on sleep (Falkenberg et al., 2021), total protein intake from meals were standardized to 1.2 g/kg bodyweight, as this is the lowest end of protein recommendations for athletes (Thomas et al., 2016). The provided dinner meal was lower in protein (0.2 g/kg) and separated from the supplement by at least 2 hr. The 40 g presleep protein supplementation was in addition to the 1.2 g/kg standardization, which resulted in a mean total protein intake of 1.7 ± 0.1 g/kg (habitual intake: 1.6 ± 0.5 g/kg).

Prior to the dinner meal, participants were permitted to consume only approved low-protein snacks external to provided foods, given they were reported and replicated in the corresponding crossover period. Due to the inhibitory action on athlete sleep (Barnard et al., 2022), caffeine consumption was not permitted after 12:00 p.m., with timing and amount replicated for each trial.

Female Participants

Females additionally completed a menstrual cycle questionnaire to determine use of contraceptive devices and menstrual status. To limit the impact of female sex hormones on sleep outcomes (Baker & Lee, 2018), naturally menstruating females ($n = 9$) completed the intervention phases during the early follicular phase (i.e., menses to Day 7) and those using monophasic oral contraceptives ($n = 2$) during the time of the active pill. Scheduling of female participants was confirmed via the use of menstrual diaries and menses (Knowles et al., 2019), with the washout period extended to ensure testing was completed during the required menstrual phase.

Sleep Assessment

Participants wore a portable dry-EEG headband (DREEM 3, DREEM) for three consecutive nights in each intervention period. The DREEM 3 headband detects five physiological signals via three embedded sensors: (a) brain cortical activity via five electroencephalogram dry electrodes yielding seven derivations (FpZ-O1, FpZ-O2, FpZ-F7, F8-F7, F7-O1, F8-O2, FpZ-F8; 250 Hz with a 0.4–35 Hz band-pass filter); (b) movements, position, and respiratory rate via a 3D accelerometer over the head; and (c) heart rate through a red-infrared pulse oximeter within the frontal band. DREEM demonstrates valid EEG signaling compared with polysomnography, with the automated DREEM algorithm accurately determining real-time sleep staging (84%) in line with the Academy of Sleep Medicine Guidelines (Arnal et al., 2020). The sleep recording was remotely started by the researcher only once these conditions were met: (a) a strong signal quality was detected, and (b) the participant was positioned in bed ready to attempt sleep. Participants were awoken at prescribed wake times, with the sleep recording stopped and data uploaded to the DREEM portal immediately.

Data were recorded in 30-s epochs and analyzed using DREEM's automated algorithm. All recordings >70% record quality were reported and included within statistical analyses, with recordings ≤70% quality removed, in line with previous research (Mallender & Depner, 2023).

Subjective Measures

Karolinska Sleepiness Scale

Sleepiness was assessed using the KSS, a Likert-type scale where participants circle a number corresponding to their level of sleepiness within the preceding 5 min, ranging from 1 (*extremely alert*) to 9 (*very sleepy, great effort keeping awake, fighting sleep*) (Miley et al., 2016). The KSS displays strong correlation with behavioral and EEG indicators of sleepiness (Kaida et al., 2006).

BRUMS

Mood was assessed using the BRUMS—a shortened version of the Profile of Mood States (Terry et al., 1999). Twenty-four mood-based adjectives are rated on a 5-point Likert scale in real time, ranging from *not at all* to *extremely*. Mood is then divided into six mood states (i.e., anger, confusion, depression, fatigue, tension, and vigor), with scoring as per original criteria (Terry et al., 1999). This shortened mood scale displays moderate–high internal consistency, with Cronbach's alpha ranging between .75 and .86 in young athletes (Terry et al., 1999).

Short Recovery Stress Scale for Sport

Subjective stress and recovery were measured using the SRSS, which contains 8 scales individually rated from 0 (*does not apply at all*) to 6 (*fully applies*) in real time (Kellmann & Kölling, 2019). Four scales pertain to the recovery aspect of the SRSS (i.e., *Physical Performance Capability*, *Mental Performance Capability*, *Emotional Balance*, and *Overall Recovery*), and the remaining 4 scales represent current stress state (i.e., *Muscular Stress*, *Lack of Activation*, *Negative Emotional State*, and *Overall Stress*). This scale has been previously validated within Australian sporting populations (Kölling et al., 2020).

Consensus Sleep Diary

The Consensus Sleep Diary is a standardized sleep tool used to record subjective sleep metrics, including sleep/wake times, sleep latency, awakenings, and sleep quality using a 5-point Likert scale (Carney et al., 2012).

Training Diary

A daily training diary was completed each evening, detailing exercise type, start/finish time, and rating of perceived exertion on a 0–10 Likert scale. Though training sessions were habitual, participants were to replicate their training sessions as close to possible each intervention period, which was confirmed by the researcher.

Cognitive Performance

Cognitive testing was conducted using the Joggle Research automated cognitive testing battery (Joggle Research). Four tests that took approximately 7 min were administered to assess cognitive components relating to sports performance susceptible to sleep

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loss (Fullagar et al., 2015). The cognitive testing battery was administered following the 30-min postwaking morning surveys to reduce the impacts of sleep inertia (Hilditch & McHill, 2019). In sequential order along with the cognitive domain assessed, tests included the *Motor Praxis task:* sensory motor speed; *Line Orientation* Task: spatial orientation; *Digital Symbol Substitution Task*: complex scanning and visual tracking; and *Psychomotor Vigilance Task*: vigilant attention (Basner et al., 2015). Primary outcomes across the tests included reaction time, standardized speed/efficiency scores, accuracy, lapses (responses > 355 ms), and false starts.

Biochemical Analysis

Whole blood samples were collected into 6 ml Vacutainers containing ethylenediaminetetraacetic acid and were centrifuged at 3,000 revolutions per minute for 10 min at 4 °C following collection of all samples. Separated plasma was pipetted into duplicate 2.5 ml tubes and stored at −80 °C.

Prior to derivatization, 100 µl aliquots of each plasma sample was added to 200 µl of acetonitrile then mixed and centrifuged at 13,200 revolutions per minute for 20 min at 4 °C.

Amine derivatization was performed using AccQ-Tag Ultra Derivatization Kit (Waters), as previously described by Boughton et al. (2011). Borate buffer containing 2.5 µM standards of L-Tryptophan-(indole-d_5) and L-Valine-($^{13}C_5$, ^{15}N) as internal standards was, however, substituted for the standard borate buffer. Post amino acid derivatization, samples were quantified using ultraperformance liquid chromatography with triple–quadrupole mass spectrometry (UPLC-QqQ-MS), as per previous methodology (Boughton et al., 2011). Changes in the TRP:LNAA ratio were determined by dividing plasma TRP concentration by the sum of the other LNAAs, that is, isoleucine, leucine, phenylalanine, tyrosine, and valine.

Statistical Analysis

Researchers remained blinded to treatment throughout statistical analyses. Data were checked for normal distribution and analyzed using restricted maximum likelihood mixed models within StataBE 18 (StataCorp LLC). The effect of dietary intervention (i.e., control or α-lactalbumin) was fitted as a fixed effect to determine whether there was a difference in the effect of dietary intervention on dependent sleep variables and cognition between treatments, with timepoint (i.e., 0, 30, 60, 90, and 120 min postsupplementation) added as a fixed effect in models determining treatment and time interactions for mood, sleepiness, recovery, and plasma amino acid concentrations. Participant identification numbers were included as a random factor to account for repeated measures within each model, with the repeated degrees of freedom method utilized. Bonferroni post hoc analyses were completed to determine significance between timepoints of each treatment. All variables are reported as mean ± *SD* with significance level set at $p < .05$.

Due to the cannulation of participants 2.5 hr prior to sleep and confounding "first night effect" (Agnew et al., 1966; Benloucif et al., 2008), the first night of each intervention period served as an adaptation night and was not included in statistical analysis (Halson et al., 2020). Therefore, residual first morning measures of sleepiness, mood, and cognition were also omitted. Data for Nights 2 and 3 were pooled together to analyze treatment effects.

Results

Blood samples were unable to be drawn from one participant for both intervention periods ($n = 18$), with their remaining data included in all other statistical analyses. Macronutrient (CHO: 3.4 ± 0.9; FAT: 1.3 ± 0.4; PRO: 1.7 ± 0.1 g/kg) and calorie consumption (2749.0 ± 499.4 kCal) was statistically similar between trials ($p > .05$).

Plasma TRP and TRP:LNAA ratio

There was a positive time and treatment interaction within the ALAC treatment. Compared with the 0-min timepoint, plasma TRP concentration and TRP:LNAA ratio was increased at the 30-, 60-, 90-, and 120-min timepoints post-ALAC consumption. Plasma levels of TRP (255.8 ± 57.9 µmol/L) and TRP:LNAA ratio (0.28 ± 0.03) were highest at 120 min post-ALAC consumption (Figure 2A and 2B).

Objective Sleep

Eighty-eight percent of DREEM sleep recordings were >70% quality and included in analyses ($n = 67$). Compared with control, NREM 2 duration ($p = .02$) and percentage ($p = .02$) was increased in the ALAC condition, whereas rapid eye movement (REM) duration ($p < .01$) and percentage ($p < .01$) were decreased (Table 3). No treatment effects were observed for any other objective sleep metrics.

Subjective Measures

Sleep

There were no significant treatment by time interactions on any subjective sleep measures from the sleep diary; however, subjective sleep quality was decreased in the ALAC condition compared with control (CON 3.7 ± 0.6, ALAC 3.3 ± 0.9; $p = .03$).

Sleepiness

There was a significant time effect on sleepiness independent of treatment, with all timepoints in the evening (i.e., 30, 60, 90, and 120 min postsupplement) and morning significantly greater than the 0-min timepoint ($p < .01$). There was an effect of treatment on the 30-min timepoint (CON 5.1 ± 1.8, ALAC 4.6 ± 1.5; $p = .013$), with no further effect of treatment across timepoints ($p > .05$).

Mood

Independent of treatment, reductions in vigor and total mood score, and significant increases in fatigue were observed for all evening timepoints compared with 0-min postsupplementation ($p < .01$), with morning measures of vigor and mood disturbance scores also increased compared with the 0-min timepoint ($p < .05$). A treatment effect was observed for tension at 120 min (CON 0.3 ± 0.8, ALAC 0.7 ± 1.1; $p = .021$); however, no treatment effects were observed between any other mood state across all timepoints.

Recovery and Stress

There were no treatments by time interactions for all measures within the SRSS. A treatment effect for muscular stress (CON 1.9 ± 1.3, ALAC 2.3 ± 1.4; $p = .02$) and overall stress in the

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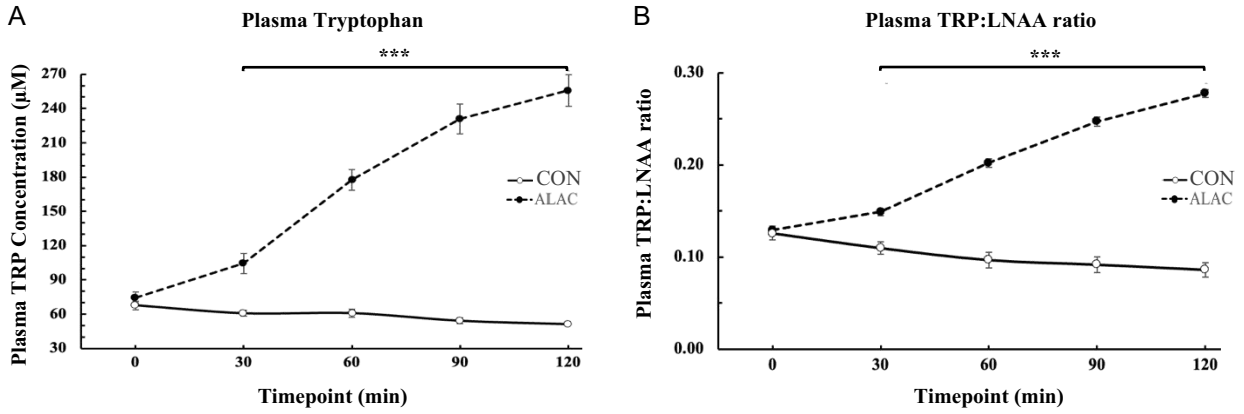

Figure 2 — Plasma concentrations of amino acids on Night 1 of each trial. (A) Plasma TRP concentrations. (B) Plasma TRP:LNAA ratio. ***Significant effect of treatment and time ($p < .01$). TRP = tryptophan; LNAA = large neutral amino acids; CON = control; ALAC = α-lactalbumin.

Table 3 Objective Sleep Metrics

Sleep metric	CON	ALAC	*p*	Treatment coefficient	95% CI
TST (min)	455.8 ± 23.8	456.6 ± 21.6	.966	0.15	[−6.76, 7.05]
WASO (min)	14.3 ± 7.2	17.1 ± 15.4	.231	2.60	[−1.70, 6.90]
SE (%)	92.8 ± 3.8	92.6 ± 4.4	.459	−0.41	[−1.50, 0.69]
SOL (min)	17.6 ± 11.9	15.5 ± 14.8	.615	−1.21	[−5.99, 3.58]
NREM Stage 1 (min)	25.2 ± 8.4	27.5 ± 8.3	.158	1.66	[−0.67, 3.99]
NREM Stage 1 (%)	5.5 ± 1.8	6.0 ± 1.9	.119	0.41	[−0.11, 0.93]
NREM Stage 2 (min)	205.9 ± 33.3	216.5 ± 33.1*	.021*	10.93	[1.72, 20.13]
NREM Stage 2 (%)	45.2 ± 7.2	47.5 ± 7.6*	.015*	2.43	[0.50, 4.36]
NREM Stage 3 (min)	114.0 ± 31.2	112.9 ± 29.5	.750	−1.36	[−9.91, 7.19]
NREM Stage 3 (%)	25.1 ± 6.7	24.7 ± 6.0	.642	−0.43	[−2.27, 1.42]
REM (min)	110.8 ± 27.9	99.7 ± 23.1*	.010*	−11.19	[−19.60, −2.79]
REM (%)	24.2 ± 5.7	21.8 ± 4.8*	.011*	−2.40	[−4.22, −0.58]

Note. Data used for analysis are the mean of Nights 2 and 3. ALAC = α-lactalbumin; CI = confidence interval; CON = control; NREM = nonrapid eye movement; REM = rapid eye movement; TST = total sleep time; SE = sleep efficiency; SOL = sleep onset latency; WASO = wake after sleep onset.
*Significantly different compared with control ($p < .05$)

morning (CON 1.4 ± 1.3, ALAC 2.1 ± 1.4; $p < .01$) was observed, with no other stress or recovery measures influenced by treatment in the morning or evening.

Cognitive Performance

Across mornings, there were no treatment by time interactions for all cognitive measures. A treatment effect was observed, with reductions in reaction time for the Motor Praxis task ($p = .02$), Line Orientation Task ($p = .03$), and Psychomotor Vigilance Task ($p = .02$) and an increase in standardized efficiency score in the Psychomotor Vigilance Task ($p = .01$) recorded during the ALAC treatment compared with control (Table 4). No significance was observed for any remaining cognitive outcomes.

Discussion

The aims of this study were to determine the effects of evening supplementation of 40 g α-lactalbumin (1.9 g TRP) on the sleep, cognition, mood, and plasma amino acid concentrations of a trained population with sleep difficulties. Compared with the control, α-lactalbumin treatment increased NREM Stage 2 duration and percentage, decreased REM duration and percentage, and improved morning cognitive function (i.e., reaction time). As such, evening α-lactalbumin supplementation may assist athletes with sleep difficulties in enhancing next-day cognitive performance, especially in the morning.

This study demonstrates that 40 g of α-lactalbumin significantly increases the plasma TRP:LNAA ratio, with TRP concentrations being highest 120 min postsupplementation. This shows that the intervention was effective, with the timing of α-lactalbumin supplementation appearing suitable 2 hr presleep. In the α-lactalbumin trial, NREM Stage 2 duration increased, on average, by 10.6 min—or 2.3% as a proportion of total sleep—compared with control. Previously, 16 trained females supplemented with 40 g α-lactalbumin in the evening were found to increase NREM Stage 2 percentage from 45.3 to 48.0% (Miles et al., 2021); a similar increase to the 45.2%–47.5% observed in this study. Subsequent REM sleep proportion decreased to accommodate the increase in NREM Stage 2 duration; however, REM sleep percentage remained

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Table 4 Cognitive Outcomes

Joggle outcome	CON	ALAC	*p*	Treatment coefficient	95% CI
Motor Praxis Task					
Reaction time (ms)	460.2 ± 61.3	440.9 ± 50.2*	.021*	−19.29	[−35.56, −3.03]
Speed score	994.1 ± 3.7	995.3 ± 2.2	.017*	1.24	[0.23, 2.24]
Line Orientation Task					
Reaction time (ms)	5,822.7 ± 2,452.2	5,008.0 ± 1,833.3*	.026*	−814.69	[−1,529.06, −100.32]
Correct responses	5.0 ± 2.1	5.0 ± 2.3	1.00	−0.00	[−0.80, 0.80]
Efficiency score	727.7 ± 106.5	754.3 ± 97.4	.113	25.58	[−6.46, 59.61]
Digital Symbol Substitution Task					
Reaction time (ms)	956.2 ± 115.8	963.2 ± 134.6	.553	6.92	[−16.32, 30.17]
Correct responses	85.1 ± 9.0	85.0 ± 10.1	.868	−0.16	[−2.05, 1.74]
Incorrect responses	0.4 ± 0.8	0.4 ± 0.8	.871	0.03	[−0.30, 0.35]
Efficiency score	951.8 ± 64.9	943.4 ± 81.9	.198	−8.39	[−21.31, 4.52]
Psychomotor Vigilance Task					
Reaction time (ms)	278.5 ± 33.2	267.1 ± 33.5*	.020*	−11.35	[−20.84, −1.86]
Efficiency score	590.1 ± 176.1	649.5 ± 194.5*	.013*	59.45	[12.86, 106.03]
Lapses (>355 ms)	4.8 ± 4.2	4.2 ± 5.1	.295	−0.63	[−1.83, 0.56]
Total errors	6.0 ± 4.2	6.1 ± 5.6	.940	0.05	[−1.34, 1.45]

Note. Data used for analysis are the mean of Days 3 and 4. ALAC = α-lactalbumin; CI = confidence interval; CON = control.
*Significantly different compared with control ($p < .05$)

within a healthy, typical range of 21%–30% (21.8 ± 4.8%; Ohayon et al., 2017). NREM Stage 2 sleep is categorized by the presence of sleep spindles (i.e., bursts of neuronal firing) and/or K-complexes (i.e., long delta waves), believed essential for brain plasticity, memory consolidation, and motor learning (Walker, 2009). Therefore, increased NREM Stage 2 sleep following α-lactalbumin supplementation may support the continual learning and cognitive adaptation that optimizes athletic performance; however, more research is required to understand the impact of chronic α-lactalbumin supplementation on sleep architecture.

Sleep latency and efficiency remained unaffected by treatment, contrary to the hypothesis that sleep difficulties would be reduced following evening α-lactalbumin supplementation. Recruited participants reported mild or greater sleep difficulty and had a mean subjective sleep latency of >30 min; however, the median objective sleep latency throughout the control treatment was 13 min—indicating that sleep onset difficulties were likely overstated or reduced within the laboratory environment (Valko et al., 2021). While evening collagen supplementation has not been seen to affect sleep latency or efficiency (Thomas et al., 2024), due to the high glycine content, it is possible that collagen supplementation could have had some effect on sleep outcomes. Nonetheless, given the blinded crossover nature of this study and the matching of supplements (i.e., energy and protein), this does not undermine the key findings, namely, that α-lactalbumin appears to affect NREM Stage 2 and REM sleep duration and improve next-day cognitive performance.

Moreover, TRP-based supplementation may elicit a loading effect, with one study of sleep onset insomniacs observing "delayed" sleep benefits, with at least three nights of pure TRP supplementation (3 g) required before sleep latency was reduced (Spinweber, 1986). It remains unknown whether α-lactalbumin may exert a similar TRP loading effect, whereby future studies should supplement for more than three nights to observe this potential effect.

Morning cognitive performance, particularly reaction time, improved following presleep α-lactalbumin supplementation. Compared with control, α-lactalbumin significantly improved reaction time in three of the four tasks within the cognitive battery, including tasks involved with sensory motor speed, spatial orientation, and vigilant attention (Basner et al., 2015). Within sports, reaction time and anticipatory skills are advantageous to player performance, responding to dynamic changes in environment (Nuri et al., 2013). Similar improvements have been observed previously, whereby morning attention and vigilance performance were increased within a cohort of poor sleepers supplemented with 2 × 20 g α-lactalbumin the previous evening (Markus et al., 2005). These benefits were purported to be through potential improvements in sleep as morning sleepiness was also reduced (Markus et al., 2005); however, within our study, objective sleep metrics, morning fatigue, vigor, and sleepiness were unaffected following α-lactalbumin supplementation. Interestingly, subjective sleep quality was slightly reduced (0.4 units) in the α-lactalbumin condition; however, this is not shown to affect working memory, executive functions, or procedural learning (Zavecz et al., 2020).

Enhanced cognitive functioning observed in this study may relate to augmented neural serotonergic activity, glutaminergic neurotransmission, and increased NREM Stage 2 sleep following TRP loading (Silber & Schmitt, 2010; Walker, 2009). A 30%–45% increase in the plasma TRP:LNAA ratio has previously shown to increase serotonin-mediated neuroendocrine changes (Markus, 2008), with the plasma TRP:LNAA ratio increasing by 87% post α-lactalbumin consumption in this current study. In addition, changes in NREM Stage 2 sleep duration following α-lactalbumin supplementation may act as a potential modulator in improved cognitive performance due to associations with cognitive adaptation (Walker, 2009). However, further research is required to understand these mechanisms further and investigate whether increased NREM sleep relates to this next-day cognitive improvement. It also remains to be seen whether reaction time improvements following evening α-lactalbumin

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supplementation are sustained next-day, or whether these benefits are limited to the morning.

This study displays robust experimental control (i.e., strict dietary standardization), with confirmation that 40 g α-lactalbumin induced meaningful changes in plasma TRP levels. While this study exhibited strong controls, limitations exist. To reduce burden, participants were permitted to complete their usual activities during the day, given they returned to the facility prior to the timing of the supplement. Though there were no differences observed in the primary variable of interest (sleep latency) within a trained population with sleep difficulties; potential confounders were limited through implementing a crossover design, completing each intervention period during the same meteorological season and scheduling (e.g., exercise and work), reporting of additional food intake via a smartphone application, and supervised intake of the supplement at the prescribed time (100% adherence). Further, we acknowledge that collagen may influence sleep through mechanisms outside of the TRP:LNAA ratio that warrants further research; however, with the current evidence-base, this control supplement best accounts for several known factors that influence sleep through this ratio (i.e., energy and protein intake).

Conclusions

This is the first study to investigate the effect of 40 g α-lactalbumin supplementation in the evening on the sleep, mood, cognition, and plasma amino acid concentration of a trained population with sleep difficulties. Presleep α-lactalbumin supplementation significantly increased NREM Stage 2 sleep, improved morning cognition, and augmented the TRP:LNAA ratio compared with control. Though sleep latency and efficiency were unchanged, acute evening supplementation of α-lactalbumin may prove beneficial to trained individuals with sleep difficulties looking to support next-day cognitive performance. Future research should test the ecological validity of subchronic (>3 nights) α-lactalbumin supplementation within a cohort of athletic individuals with sleep difficulties.

Practical Applications

- Consuming 40 g α-lactalbumin 2 hr before bedtime may increase NREM Stage 2 sleep, cognitive efficiency, and reduce reaction time in athletes with sleep difficulties.
- Separate α-lactalbumin supplementation from a low-protein (0.2 g/kg bodyweight) evening meal by ~2 hr to maximize the TRP:LNAA ratio response.
- Evening supplementation of 40 g α-lactalbumin does not impair objective sleep.

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Nontechnical Summary

Poor sleep is common among athletes and active populations due to factors such as training load and competition. Chronic poor sleep influences the recovery, performance, and overall wellness of an athlete; therefore, more practical strategies are required to reduce the risk of poor sleep in these individuals.

Recently, diet has been seen to influence the sleep of an athlete, particularly protein intake in the evening. Protein intake high in the amino acid tryptophan is of particular interest, given that tryptophan serves as a precursor to serotonin and melatonin production—two primary hormones that promote sleep maintenance and initiation. This is of particular interest to athletes, given whey protein is high in tryptophan and is commonly consumed in the evening. The highest protein source of tryptophan, α-lactalbumin (whey protein fraction), has been seen to increase sleepiness in the evening yet increase morning alertness following a more restful night of sleep. Therefore, this research investigated the influence of α-lactalbumin on the sleep, sleepiness, mood, and next-morning cognitive performance of an athletic population that reports having poor sleep.

Nineteen athletically trained individuals participated in the study, all of which slept poorly as identified by scoring ≥5 on two different sleep screening questionnaires (Athlete Sleep Screening Questionnaire and Pittsburgh Sleep Quality Index). Participants received either 40 g α-lactalbumin or an isocaloric control (collagen) 2 hr before bedtime in a controlled environment for three nights, in a double-blinded, randomized, crossover trial. Sleep was measured using a DREEM headband across all nights (portable EEG), with blood samples taken only on the first night of each experimental trial to measure levels of tryptophan and other amino acids within the blood postsupplement consumption. Mood, sleepiness, and recovery questionnaires were also assessed each morning and night. Following sleep, each morning a cognitive testing battery (four tasks) was completed on an iPad to assess different cognitive domains sensitive to sleep changes.

The results showed that during the α-lactalbumin trial compared with the control, (a) tryptophan levels in the blood were increased. (b) NREM Stage 2 sleep increased, and REM sleep decreased. (c) Morning cognitive performance was improved (three tasks had significantly quicker reaction times).

These results suggest that athletes reporting poor sleep quality may benefit from α-lactalbumin supplementation to assist with their next-day performance. As the direct mechanism by which this resultant increase in cognitive performance occurs is not well understood, more research should aim to test this concept. Future research should explore this concept in a similar population outside of a tightly controlled setting and observe whether next-day sporting performance is influenced through α-lactalbumin supplementation.

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